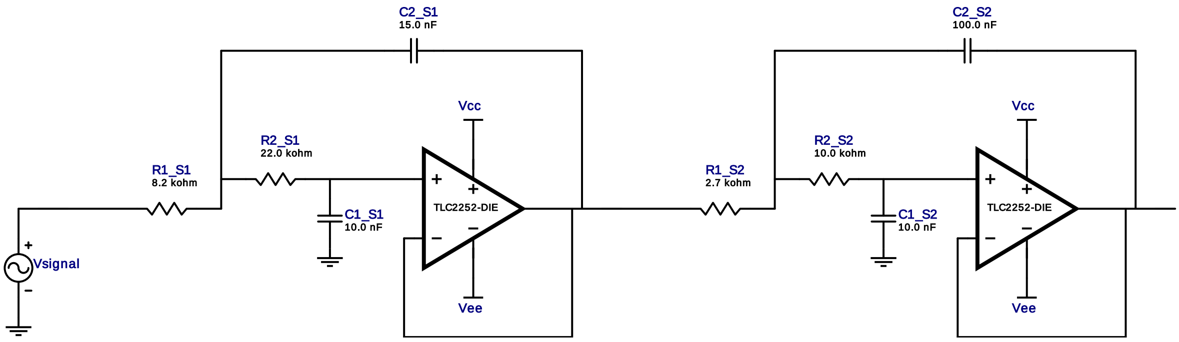
Type : Lowpass
Response : Butterworth
Order : 4
Number of Stages : 2

Filter Design Report

Design : Lowpass Filter - 4th order Butterworth
Design ID: 5

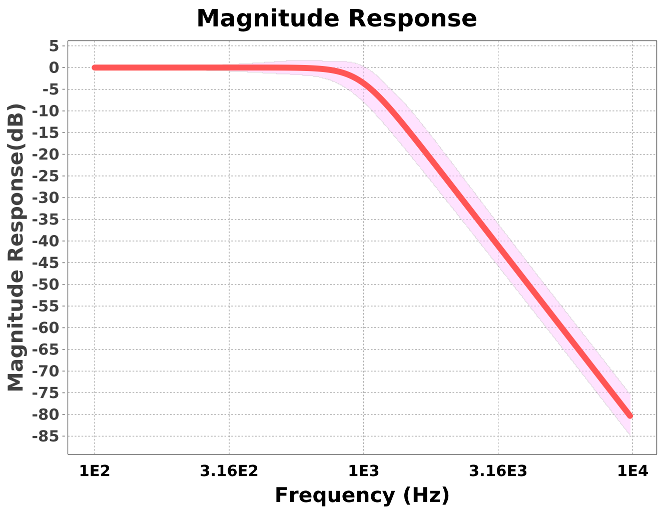


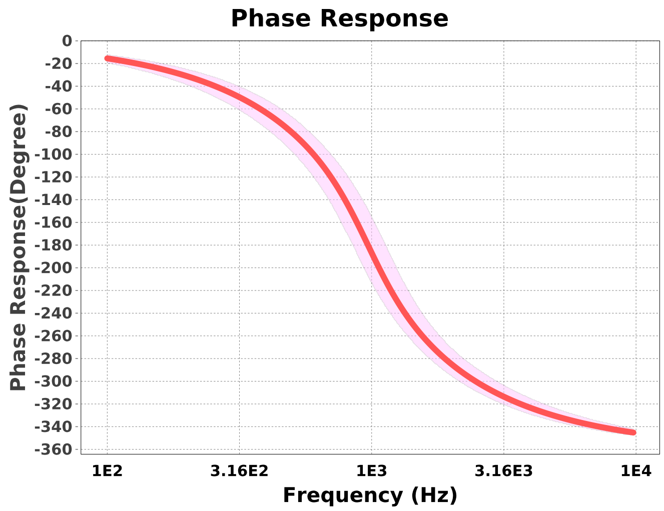
Electrical BOM

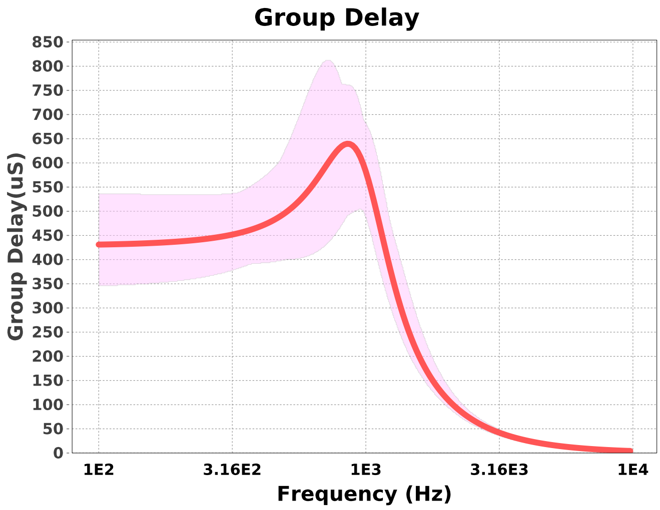
#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	TLC2252-DIE	GbwTyp= 0.2MHz VccMax= 16V VccMin= 4.4V	1
2.	A1_S2	Texas Instruments Inc.	TLC2252-DIE	GbwTyp= 0.2MHz VccMax= 16V VccMin= 4.4V	1
3.	C1_S1	Generic	Ideal	Cap= 10.0 nF Tolerance= 20.0 %	1
4.	C1_S2	Generic	Ideal	Cap= 10.0 nF Tolerance= 20.0 %	1
5.	C2_S1	Generic	Ideal	Cap= 15.0 nF Tolerance= 20.0 %	1
6.	C2_S2	Generic	Ideal	Cap= 100.0 nF Tolerance= 20.0 %	1
7.	R1_S1	Generic	Ideal	Res= 8200.0ohm Tolerance= 10%	1
8.	R1_S2	Generic	Ideal	Res= 2700.0ohm Tolerance= 10%	1
9.	R2_S1	Generic	Ideal	Res= 22000.0ohm Tolerance= 10%	1
10.	R2_S2	Generic	Ideal	Res= 10000.0ohm Tolerance= 10%	1

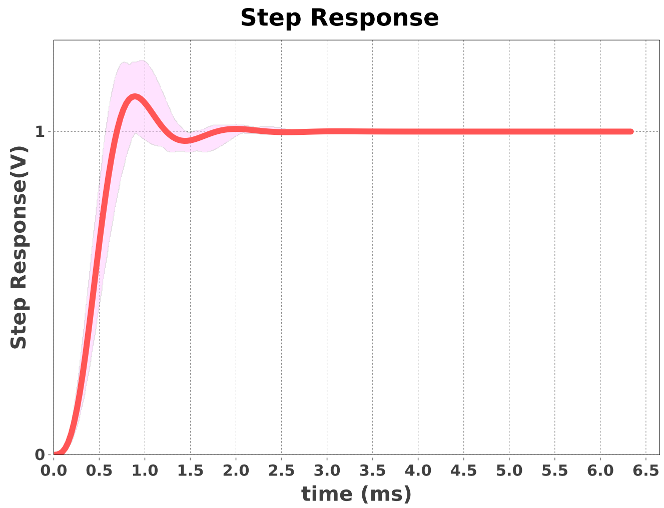
Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E6	20%
2.	Res	E12	10%









Design Inputs

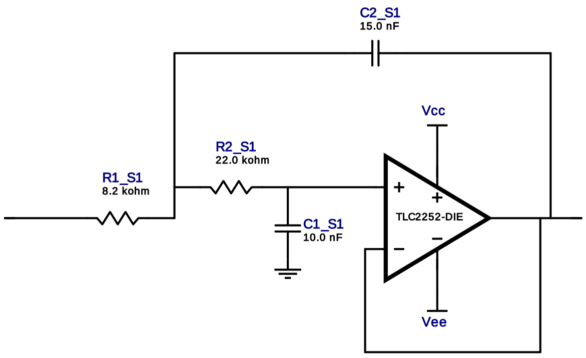
#	Name	Value	Description
1.	FilterType	lowpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	4.0	
4.	FilterTopology	Sallen-Key	
5.	NumberOfStages	2.0	
6.	PassbandFrequency	1,000.0	
7.	StopbandAttenuation	-79.999	
8.	StopbandFrequency	10.0 k	
9.	Gain	1.0	
10.	DualSupply	+/-5.00 V	Power supply(s) to active chips
11.	ResistorTolerance	E12	Resistor series - 10% Passive resistor tolerance
12.	CapacitorTolerance	E6	Capacitor series - 20% Passive capacitor tolerance

Design Assistance

1. **TLC2252-DIE** Product Folder : http://www.ti.com/product/TLC2252-DIE : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency	967.512 Hz
Min GBW Reqd	54.121 kHz
Stage Gain	1.0 V/V
Stage Q	544.699 m
Stage Topology	Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	TLC2252-DIE	GbwTyp= 0.2MHz VccMax= 16V VccMin= 4.4V	1
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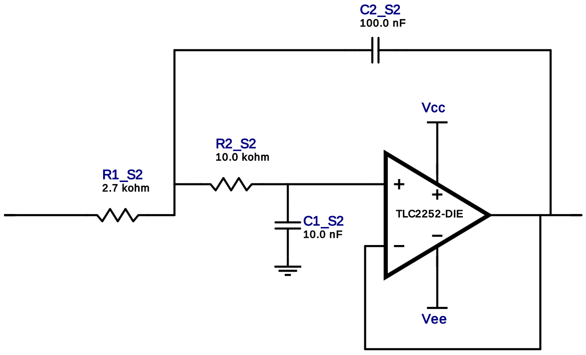
3.	C2_S1	Generic	Ideal	Cap= 15.0 nF Tolerance= 20.0 %	1
4.	R1_S1	Generic	Ideal	Res= 8200.0ohm Tolerance= 10%	1
5.	R2_S1	Generic	Ideal	Res= 22000.0ohm Tolerance= 10%	1

Filter Stage :2

Cutoff Frequency	968.586 Hz
Min GBW Reqd	130.657 kHz
Stage Gain	1.0 V/V
Stage Q	1.294
Stage Topology	Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	TLC2252-DIE	GbwTyp= 0.2MHz VccMax= 16V VccMin= 4.4V	1
2.	C1_S2	Generic	Ideal	Cap= 10.0 nF Tolerance= 20.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 100.0 nF Tolerance= 20.0 %	1
4.	R1_S2	Generic	Ideal	Res= 2700.0ohm Tolerance= 10%	1
5.	R2_S2	Generic	Ideal	Res= 10000.0ohm Tolerance= 10%	1